

NovaDiff — Change report

Baseline: f7c2bfd81af2

Target: NovaDiff

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Diagrams use Mermaid heuristics (imports, cross-file call guesses, type sketches). They are **not** compiler-grade call graphs. The module graph is file-to-file (JS/TS relative imports). The call graph links files when a callee name maps to exactly one definition in the repo (regex across many languages).

AI project narrative

Workspace overview — roots, comparison intent, scale of change

The diff compares a snapshot stored under

`/Users/iamgroot/Library/Application Support/novadiff/novadiff-workspaces/7d1b19ef-9547-48e8-92d5-c8beabe3849c/snapshots/f7c2bfd81af2`

with the current repository at

`/Users/iamgroot/Documents/GitHub/NovaDiff`.

A total of **62** paths were touched: **35 modified**, **23 added**, **4 removed**.

The majority of changes cluster in the `src/`, `electron/`, and `packages/graph-view/` directories, indicating a focus on UI, Electron integration, and graph-view logic.

Change landscape — interpret counts, dominant extensions, depth hotspots, risk intuition

- **File types**
 - `.tsx` (24) and `.ts` (17) dominate, reflecting UI and core logic updates.
 - `.cjs` (13) shows new CommonJS modules for Electron.
 - `.json` and `.css` changes are minimal.
- **Path depth**
 - 27 paths at depth 3, 9 at depth 4, 4 at depth 5.
 - Depth-4 changes are mainly in `electron/` and `packages/graph-view/`.
- **Risk signals**
 - `vite.config.ts` was modified (deterministic risk: medium).
 - Removal of `.novadiff-graph/config.json`, `diff-overlay.json`, and `meta.json` may break legacy graph-generation workflows.
 - New Electron modules (`export-snapshot.cjs`, `git-commit.cjs`, `git-publish.cjs`) introduce runtime dependencies on Git and GitHub, raising security and authentication concerns.

Subsystem map — group paths into coherent areas

- **Electron core** (`electron/`)
 - New modules: `export-snapshot.cjs`, `git-commit.cjs`, `git-publish.cjs`.
 - Updated `git-service.cjs` and `github-service.cjs` to support staged paths.
- **Graph view UI** (`packages/graph-view/src/`)
 - Updated explorer components (`NovaDiffGraphExplorer.tsx`, `NovaDiffGraphExplorerEmbed.tsx`).
 - Added selection clustering utilities (`selectionCluster.ts`) and tests (`selectionCluster.test.ts`).
- **Application logic** (`src/`)
 - Commit analysis (`commitChangeAnalysis.ts`), file tree (`fileTree.ts`), workspace storage (`workspaceStorage.ts`).
 - New UI helpers: `DocsViewSyncContext.tsx`, `DocumentationWorkspace.tsx`.
- **Configuration & tooling**
 - `vite.config.ts` change.
 - Removal of legacy `.novadiff-graph` config files.
 - New `novadiff-docs-html.cjs` for HTML rendering of docs.
- **Testing**
 - Single test file `graphCityBridge.test.ts` added.

Cross-cutting concerns — security, build/release, migrations, observability

- **Security**
 - GitHub authentication flows (`github-auth-flow.cjs`, `github-commit-context.cjs`) now exposed in the main process; ensure tokens are not logged.

- LLM module (`llm.cjs`) may send user code to external services; verify compliance with privacy policies.
- **Build/Release**
 - `vite.config.ts` alteration could affect asset bundling; run a full build test.
 - Electron packaging must include new `.cjs` modules; update `package.json` scripts accordingly.
- **Migrations**
 - Deletion of `.novadiff-graph/config.json` and related files requires updating any scripts that load graph configuration.
 - Existing snapshots may need regeneration to include new snapshot export format.
- **Observability**
 - New modules lack explicit logging; consider adding console traces for debugging.
 - No new metrics are introduced, but the LLM module may benefit from response-time monitoring.

Documentation & tooling gaps — what would require Doxygen/clangd/tree-sitter or runtime profiling to validate

- **CJS modules:** Added CommonJS files lack type declarations; static analysis tools (e.g., TypeScript's `--allowJs`) could flag missing exports.
- **LLM integration:** Runtime profiling is needed to measure latency and token usage; no compile-time checks exist.
- **Graph view utilities:** The selection clustering logic (`utils/selectionCluster.ts`) is complex; unit tests cover basic cases but deeper integration tests would confirm correctness.
- **GitHub API usage:** Mocking of GitHub responses is not present; integration tests should verify error handling for rate limits.

Suggested verification — tests, manual checks, staged rollout

- **Unit tests**
 - Run `npm test` to cover new `selectionCluster.test.ts` and existing tests.
 - Add tests for `git-publish.cjs` staged-path logic.
- **Integration tests**
 - Spin up a local Electron instance to exercise snapshot export and commit detail retrieval.
 - Mock GitHub responses to validate authentication flows.
- **Manual checks**
 - Verify that the UI still renders the graph view correctly after the component updates.
 - Confirm that the `vite` build produces the expected assets.
- **Staged rollout**
 - Deploy to a staging environment first; monitor LLM response times and GitHub API usage.
 - Use feature flags for the new snapshot export to enable rollback if issues arise.

Compare metrics

Compare metrics

Metric Count

Added 23

Removed 4

Modified 35

Baseline: f7c2bfd81af2

Target: NovaDiff

Changed paths (sample, up to 80)

- **removed** .novadiff-graph/config.json
- **removed** .novadiff-graph/diff-overlay.json
- **removed** .novadiff-graph/knowledge-graph.json
- **removed** .novadiff-graph/meta.json
- **added** electron/export-snapshot.cjs
- **added** electron/git-commit.cjs
- **modified** electron/git-publish.cjs
- **modified** electron/git-service.cjs
- **added** electron/github-commit-context.cjs
- **modified** electron/github-service.cjs
- **modified** electron/llm.cjs
- **modified** electron/main.cjs
- **modified** electron/novadiff-docs-html.cjs
- **modified** electron/prefetch-summaries.cjs
- **modified** electron/preload.cjs
- **modified** electron/workspace-history.cjs
- **modified** electron/workspace-store.cjs
- **modified** packages/graph-view/src/App.tsx
- **added** packages/graph-view/src/GraphEmbedSyncBridge.tsx
- **modified** packages/graph-view/src/NovaDiffGraphExplorer.tsx
- **modified** packages/graph-view/src/NovaDiffGraphExplorerEmbed.tsx
- **added** packages/graph-view/src/__tests__/selectionCluster.test.ts
- **modified** packages/graph-view/src/components/GraphView.tsx
- **modified** packages/graph-view/src/index.css
- **modified** packages/graph-view/src/store.ts
- **added** packages/graph-view/src/utils/graphCityNavigation.ts
- **added** packages/graph-view/src/utils/selectionCluster.ts
- **modified** src/App.css
- **modified** src/App.tsx
- **added** src/app/DocsViewSyncContext.tsx
- **added** src/app/codeCityChromeInsets.ts
- **added** src/app/commitChangeAnalysis.ts
- **modified** src/app/commitMessage.ts
- **added** src/app/fileTree.ts
- **modified** src/app/gitTypes.ts
- **added** src/app/graphCityBridge.ts
- **added** src/app/helpCopy.ts
- **added** src/app/impactHeatmap.ts
- **modified** src/app/workspaceStorage.ts
- **modified** src/app/workspaceTypes.ts
- **modified** src/components/AutoCommitWorkspace.tsx
- **modified** src/components/CodeCityLegend.tsx

- **added** src/components/CodeCityMinimapPortal.tsx
- **added** src/components/CodeCityMinimapSlot.tsx
- **modified** src/components/CodeCityView.tsx
- **added** src/components/DocsGraphExploreChrome.tsx
- **modified** src/components/DocumentationWorkspace.tsx
- **added** src/components/DocumentationWorkspaceLinked.tsx
- **added** src/components/GitBlameLineModal.tsx
- **modified** src/components/GitBlamePanel.tsx
- **added** src/components/GitHistoryContextPanel.tsx
- **modified** src/components/GitHistoryWorkspace.tsx
- **modified** src/components/HistoryCompareStrip.tsx
- **added** src/components/KeyboardShortcutsHelp.tsx
- **modified** src/components/KnowledgeGraphPanel.tsx
- **modified** src/components/SidebarNav.tsx
- **added** src/components/onboarding/ProductTour.tsx
- **modified** src/components/onboarding/onboarding.css
- **modified** src/components/ui/ui-transitions.css
- **modified** src/vite-env.d.ts
- **added** tests/graphCityBridge.test.ts
- **modified** vite.config.ts

Detected stacks (marker files)

Stack	Markers
node	package.json

Class / symbol index (heuristic)

File	Symbols / defs (regex)
cli/src/main.rs	ChangeKind, path_segments, path_depth, parent_dir, subtree_change_counts, try_build_gitignore_for_root, is_novadiff_docs_reserved_rel, filter_out_novadiff_docs_changes, walk_path_allowed, filter_changes_for_prefetch_gitignore, is_utf8_text_file, is_change_text_eligible, filter_out_non_text_changes, prefetch_summary_queue, empty_summary_evidence, compact_snippet, push_unique_string, line_range_label, detect_file_roles, build_changed_line_ranges, changed_lines_for_side, collect_touched_symbols, is_import_like_line, is_export_like_line, collect_changed_decl_lines, collect_cited_changed_lines, build_verification_hints, build_evidence_badges
electron/code-city-model.cjs	normalizeSide, dominantAuthor, ownershipForRange, selectBlamePaths, buildBlameMap, buildSymbolFileIndex, buildSideNodes, buildCodeCityModelPayload
electron/compare-runner.cjs	cliExecutableName, resolveRustCli, cleanEnv, runCompareEngine, parseEngineStdout, runCompareEngineAsync
electron/diff-excerpt.cjs	buildDiffExcerpt, buildDiffExcerptChunks
electron/export-snapshot.cjs	exportProjectSnapshotZip
electron/file-summary-export.cjs	escapeHtml, escapeMermaidForHtml, injectMermaidFences, markdownToHtml, isReservedNovadiffRel, safeSummarySegments, resolveSummaryDirAbs, safeArtifactSegment, resolveSelectionBaseDirAbs, resolveSelectionDirAbs, buildFileSummaryDocument, buildSelectionSummaryDocument, exportFileSummaryArtifacts, readFileSummaryMarkdowns, exportSelectionSummaryArtifacts, readSelectionSummaryMarkdowns
electron/gh-install.cjs	resolveBrew, getInstallPlan, readGhVersion, getGhToolingStatus, runInstallCommand, installGh
electron/gh-path.cjs	homeDir, extraPathDirs, augmentPathForCli, ghCandidatePaths, resolveViaLoginShell, verifyGh, resolveGhExecutable, clearGhCache, isGhInstalled, ghInstallHint, ghNotFoundError
electron/git-blame.cjs	parseBlamePorcelain, blameFileAtRef, blameFileOwnership
electron/git-commit.cjs	getCommitDetail
electron/git-publish.cjs	buildFullCommitMessage, previewPublish, executePublish
electron/git-service.cjs	gitExecutable, runGit, tryRunGit, isGitRepo, parsePorcelainStatus, getRepoStatus, parseGithubSlugFromUrl, stageAll, stagePaths, unstagePaths, stageDistrict, commit, push, ensureWorktreeBase, removeDirSafe, checkoutRefWorktree, cleanupNovadiffWorktrees, preparePrCompareRoots, detectGitTooling
electron/github-auth-flow.cjs	stripAnsi, cancelGithubDeviceAuth, startGithubDeviceAuth
electron/github-commit-context.cjs	parseJsonSafe, shortGhError, normalizeThread, getGithubCommitContext
electron/github-service.cjs	runGh, tryRunGh, isGhAvailable, getAuthStatus, listRepos, listPullRequests, viewPullRequest, createPullRequest, getUserProfile, listDetectedAccounts, slugFromRepoRoot
electron/graph/extract-structure.mjs	main, buildResult
electron/knowledge-graph-runner.cjs	resolveRepoRoot, resolveNodeExecutable, nodeChildEnv, runCommand, ensureGraphCoreBuilt, loadCore, isTextCandidate, walkProjectFiles, detectLanguage, fileCategory, resolveRelativeImport, buildBatchImportData, runExtractStructureBatch, mapStructuralAnalysis, heuristicComplexity, buildFileMeta, topSegment, buildLayersFromFiles, assignNodeLayers, buildDiffOverlay, getGitHash, buildKnowledgeGraph
electron/llm.cjs	normalizeBase, sendAccumulated, normalizeRepeatUnit, arraysEqual, trimRepeatedParagraphSuffix, trimRepeatedLineSuffix, sanitizeModelResponse, ollamaGenerationOptions, lmStudioGenerationOptions, ollamaNumPredict, streamOllamaChat, lmStudioMaxTokens, streamLmStudioChat, buildWorkspaceDocPrompt, buildCodebaseDocPrompt, buildCommitMessagePrompt, buildSelectionDocPrompt, bulletBlock, renderSummaryEvidenceBlock, buildExplainCodePrompt, buildGroundingContext, verifierWordLimit, buildGroundedRevisionPrompt, shouldRunVerifier, reviseGroundedMarkdown, buildPerFileSummarizePrompt, buildChunkExtractionPrompt, buildChunkSynthesisPrompt

File	Symbols / defs (regex)
electron/main.cjs	sendEngineProgress, getTargetWindow, windowStatePayload, emitWindowState, iconPath, buildMenu, createWindow, sendKnowledgeGraphProgress, sendWorkspaceHistoryProgress, sendGithubAuthProgress
electron/novadiff-docs-html.cjs	bundleKeyOf, bundleBaseDir, bundleModeMeta, escapeHtml, escapeMermaidForHtml, escapeRegex, countMatches, normalizeMalformedCodeFences, looksLikeRawDiffDump, normalizeFenceChunk, normalizeCompactListMarkers, injectMermaidFences, normalizeMarkdownDocument, stripTags, slugify, addHeadingAnchors, markdownToHtml, mermaidSection, navHtml, topDirectory, buildImpactHeatmapHtml, buildSearchIndex, buildEnhancementScript, wrapPage, bundleMetaLine, buildIndexHtml, buildNarrativeHtml, buildMetricsHtml
electron/novadiff-docs-pdf.cjs	bundleKeyOf, bundleBaseDir, bundleModeMeta, escapeHtml, escapeMermaidForHtml, renderSymbolsTable, renderImportEdgesTable, renderDetectedProjects, renderCrossFileCallEdges, mermaidBlock, buildMainPdfHtml, buildDiagramsPdfHtml, waitForPdfReady, htmlToPdfFile, generateNovadiffDocsPdf
electron/novadiff-docs-writer.cjs	bundleKeyOf, bundleBaseDir, bundleHeadline, readme, writeNovadiffDocsBundle
electron/prefetch-summaries.cjs	clearSummaryPrefetchCache, stopSummaryPrefetchWorker, getPrefetchedSummary, defaultPrefetchLimit, runSummaryPrefetchLoop, startSummaryPrefetchWorker
electron/repo-discovery.cjs	homeSearchRoots, readGitRemoteSlug, scanForGitRepos, discoverRepos, matchRepoForGithubRepo
electron/workspace-files.cjs	listSnapshotFiles, readSnapshotTextFile
electron/workspace-history.cjs	removeDirSafe, isUsableSnapshotDir, listCommitsOldestFirst, snapshotCommit, ensureCommitSnapshot, indexWorkspaceHistory, refreshWorkspaceHistory
electron/workspace-store.cjs	sessionPath, workspacesRoot, readJson, withSessionWriteLock, writeJsonAtomic, writeJson, loadSession, setLocalOnlyMode, saveSession, listWorkspaces, upsertWorkspace, setActiveWorkspace, setGitUser, getWorkspace, updateWorkspaceUiState, updateWorkspaceLiveRepo, createWorkspace
packages/graph-core/scripts/generate-alphabetical-catalog.mjs	normalizeKey, slugify, resolveWasmsKey, resolveExtensions, claimExtensions
packages/graph-core/src/__tests__/change-classifier.test.ts	makeAnalysis
packages/graph-core/src/__tests__/plugin-registry.test.ts	createMockPlugin
packages/graph-core/src/__tests__/staleness.test.ts	makeGraph
packages/graph-core/src/analyzer/graph-builder.ts	FileMeta, FileAnalysisMeta, NonCodeFileMeta, NonCodeFileAnalysisMeta, GraphBuilder
packages/graph-core/src/analyzer/language-lesson.ts	LanguageLessonResult, buildConceptPatterns, detectLanguageConcepts, getLanguageDisplayName, buildLanguageLessonPrompt, extractJson, parseLanguageLessonResponse
packages/graph-core/src/analyzer/layer-detector.ts	LLMLayerResponse, toLayerId, matchFileToLayer, detectLayers, buildLayerDetectionPrompt, parseLayerDetectionResponse, applyLLMLayers
packages/graph-core/src/analyzer/llm-analyzer.ts	LLMFileAnalysis, LLMProjectSummary, buildFileAnalysisPrompt, buildProjectSummaryPrompt, extractJson, parseFileAnalysisResponse, parseProjectSummaryResponse
packages/graph-core/src/analyzer/normalize-graph.ts	stripToValidPrefix, normalizeNodeId, normalizeComplexity, DroppedEdge, NormalizationStats, NormalizeBatchResult, inferTypeFromId, normalizeBatchOutput
packages/graph-core/src/analyzer/tour-generator.ts	buildTourGenerationPrompt, parseTourGenerationResponse, generateHeuristicTour
packages/graph-core/src/change-classifier.ts	UpdateDecision, classifyUpdate, detectDirectoryChanges, topDirectory, summarizeChanges
packages/graph-core/src/embedding-search.ts	SemanticSearchOptions, cosineSimilarity, SemanticSearchEngine
packages/graph-core/src/fingerprint.ts	FunctionFingerprint, ClassFingerprint, ImportFingerprint, FileFingerprint, FingerprintStore, FileChangeResult, ChangeAnalysis, contentHash, extractFileFingerprint, compareFingerprints, buildFingerprintStore, analyzeChanges
packages/graph-core/src/ignore-filter.ts	IgnoreFilter, createIgnoreFilter

File	Symbols / defs (regex)
packages/graph-core/src/ignore-generator.ts	parseGitIgnorePatterns, isCoveredByDefaults, generateStarterIgnoreFile
packages/graph-core/src/languages/catalog-builder.ts	catalogEntryToLanguageConfig, buildAlphabeticalLanguageConfigs, listAlphabeticalLanguages
packages/graph-core/src/languages/catalog-types.ts	LanguageCatalogEntry
packages/graph-core/src/languages/framework-registry.ts	FrameworkRegistry
packages/graph-core/src/languages/language-registry.ts	LanguageRegistry
packages/graph-core/src/languages/merge-configs.ts	mergeLanguageConfigs, languageConfigsForRegistry
packages/graph-core/src/persistence/index.ts	ensureDir, sanitiseFilePaths, saveGraph, loadGraph, saveMeta, loadMeta, saveFingerprints, loadFingerprints, saveConfig, loadConfig, saveDomainGraph, loadDomainGraph
packages/graph-core/src/plugins/discovery.ts	PluginEntry, PluginConfig, parsePluginConfig, serializePluginConfig
packages/graph-core/src/plugins/extractors/__tests__/cpp-extractor.test.ts	parse
packages/graph-core/src/plugins/extractors/__tests__/csharp-extractor.test.ts	parse
packages/graph-core/src/plugins/extractors/__tests__/go-extractor.test.ts	parse
packages/graph-core/src/plugins/extractors/__tests__/java-extractor.test.ts	parse, Repository
packages/graph-core/src/plugins/extractors/__tests__/php-extractor.test.ts	parse, helper, greet, noReturn, noop, multiline, Loggable, Repository, publicFunc, anotherFunc, myFunc, process, main, helper, findUser
packages/graph-core/src/plugins/extractors/__tests__/python-extractor.test.ts	parse
packages/graph-core/src/plugins/extractors/__tests__/ruby-extractor.test.ts	parse
packages/graph-core/src/plugins/extractors/__tests__/rust-extractor.test.ts	parse
packages/graph-core/src/plugins/extractors/base-extractor.ts	traverse, getStringValue, findChild, findChildren, hasChildOfType
packages/graph-core/src/plugins/extractors/cpp-extractor.ts	unwrapDeclaratorName, extractFuncDeclName, extractParams, extractReturnType, isStatic, CppExtractor
packages/graph-core/src/plugins/extractors/csharp-extractor.ts	extractParams, extractReturnType, hasModifier, extractUsingSource, lastComponent, extractInvocationName, CSharpExtractor
packages/graph-core/src/plugins/extractors/generic-extractor.ts	nodeName, lineRange, importSource, GenericExtractor
packages/graph-core/src/plugins/extractors/go-extractor.ts	extractParams, extractResultType, extractReceiverType, isExported, GoExtractor
packages/graph-core/src/plugins/extractors/java-extractor.ts	extractParams, extractReturnType, hasModifier, extractScopedIdentifierPath, lastComponent, JavaExtractor
packages/graph-core/src/plugins/extractors/php-extractor.ts	extractParams, extractReturnType, extractUseName, lastSegment, PhpExtractor
packages/graph-core/src/plugins/extractors/python-extractor.ts	extractParams, extractReturnType, unwrapDecorated, PythonExtractor
packages/graph-core/src/plugins/extractors/ruby-extractor.ts	extractParams, extractAttrProperties, getStringContent, RubyExtractor
packages/graph-core/src/plugins/extractors/rust-extractor.ts	extractParams, extractReturnType, isPublic, extractScopedPath, RustExtractor

File	Symbols / defs (regex)
packages/graph-core/src/plugins/extractors/types.ts	LanguageExtractor
packages/graph-core/src/plugins/extractors/typescript-extractor.ts	extractParams, extractReturnType, extractImportSpecifiers, TypeScriptExtractor
packages/graph-core/src/plugins/parsers/dockerfile-parser.ts	DockerfileParser

Module import edges (tabular)

Use together with the flowchart in the Diagrams PDF for a “call graph”-style overview at module boundaries.

From	To
electron/code-city-model.cjs	electron/git-blame.cjs
electron/file-summary-export.cjs	electron/novadiff-docs-pdf.cjs
electron/file-summary-export.cjs	electron/novadiff-docs-html.cjs
electron/gh-install.cjs	electron/gh-path.cjs
electron/git-commit.cjs	electron/git-service.cjs
electron/git-publish.cjs	electron/git-service.cjs
electron/git-publish.cjs	electron/github-service.cjs
electron/github-auth-flow.cjs	electron/gh-path.cjs
electron/github-commit-context.cjs	electron/github-service.cjs
electron/github-service.cjs	electron/git-service.cjs
electron/github-service.cjs	electron/gh-path.cjs
electron/github-service.cjs	electron/github-commit-context.cjs
electron/main.cjs	electron/compare-runner.cjs
electron/main.cjs	electron/llm.cjs
electron/main.cjs	electron/prefetch-summaries.cjs
electron/main.cjs	electron/novadiff-docs-writer.cjs
electron/main.cjs	electron/novadiff-docs-pdf.cjs
electron/main.cjs	electron/novadiff-docs-html.cjs
electron/main.cjs	electron/file-summary-export.cjs
electron/main.cjs	electron/code-city-model.cjs
electron/main.cjs	electron/knowledge-graph-runner.cjs
electron/main.cjs	electron/git-service.cjs
electron/main.cjs	electron/github-service.cjs
electron/main.cjs	electron/git-commit.cjs
electron/main.cjs	electron/repo-discovery.cjs
electron/main.cjs	electron/git-publish.cjs
electron/main.cjs	electron/workspace-store.cjs
electron/main.cjs	electron/git-blame.cjs
electron/main.cjs	electron/workspace-files.cjs
electron/main.cjs	electron/workspace-history.cjs
electron/main.cjs	electron/github-auth-flow.cjs
electron/main.cjs	electron/gh-path.cjs
electron/main.cjs	electron/gh-install.cjs
electron/main.cjs	electron/export-snapshot.cjs
electron/main.cjs	electron/compare-runner.cjs
electron/prefetch-summaries.cjs	electron/llm.cjs
electron/prefetch-summaries.cjs	electron/diff-excerpt.cjs
electron/prefetch-summaries.cjs	electron/file-summary-export.cjs
electron/prefetch-summaries.cjs	electron/git-service.cjs
electron/repo-discovery.cjs	electron/git-service.cjs
electron/workspace-history.cjs	electron/workspace-store.cjs
electron/workspace-history.cjs	electron/workspace-store.cjs
packages/graph-core/src/__tests__/ignore-filter.test.ts	packages/graph-core/src/ignore-filter.ts
packages/graph-core/src/__tests__/ignore-generator.test.ts	packages/graph-core/src/ignore-generator.ts

Cross-file call edges (tabular)

Edges appear only when a call site name uniquely resolves to a single definition path (very conservative).

From file	To file	Callee
cli/src/main.rs	electron/git-service.cjs	push
electron/code-city-model.cjs	electron/git-blame.cjs	blameFileOwnership
electron/file-summary-export.cjs	electron/novadiff-docs-html.cjs	normalizeMarkdownDocument
electron/file-summary-export.cjs	electron/novadiff-docs-pdf.cjs	htmlToPdfFile
electron/gh-install.cjs	electron/gh-path.cjs	augmentPathForCli
electron/gh-install.cjs	electron/gh-path.cjs	isGhInstalled
electron/gh-install.cjs	electron/gh-path.cjs	resolveGhExecutable
electron/gh-install.cjs	electron/gh-path.cjs	clearGhCache
electron/git-commit.cjs	electron/git-service.cjs	runGit
electron/git-commit.cjs	electron/git-service.cjs	tryRunGit
electron/git-publish.cjs	electron/git-service.cjs	isGitRepo
electron/git-publish.cjs	electron/git-service.cjs	getRepoStatus
electron/git-publish.cjs	electron/github-service.cjs	getAuthStatus
electron/git-publish.cjs	electron/github-service.cjs	slugFromRepoRoot
electron/git-publish.cjs	electron/git-service.cjs	stagePaths
electron/git-publish.cjs	electron/git-service.cjs	stageAll
electron/git-publish.cjs	electron/git-service.cjs	commit
electron/git-publish.cjs	electron/git-service.cjs	push
electron/git-publish.cjs	electron/github-service.cjs	createPullRequest
electron/github-auth-flow.cjs	electron/gh-path.cjs	isGhInstalled
electron/github-auth-flow.cjs	electron/gh-path.cjs	ghNotFoundError
electron/github-auth-flow.cjs	electron/gh-path.cjs	resolveGhExecutable
electron/github-auth-flow.cjs	electron/gh-path.cjs	augmentPathForCli
electron/github-commit-context.cjs	electron/github-service.cjs	getAuthStatus
electron/github-commit-context.cjs	electron/github-service.cjs	tryRunGh
electron/github-service.cjs	electron/gh-path.cjs	resolveGhExecutable
electron/github-service.cjs	electron/gh-path.cjs	augmentPathForCli
electron/github-service.cjs	electron/gh-path.cjs	ghNotFoundError
electron/github-service.cjs	electron/git-service.cjs	parseGithubSlugFromUrl
electron/graph/extract-structure.mjs	packages/graph-core/src/plugins/parsers/index.ts	registerAllParsers
electron/knowledge-graph-runner.cjs	packages/graph-core/src/analyzer/graph-builder.ts	GraphBuilder
electron/knowledge-graph-runner.cjs	packages/graph-core/src/persistence/index.ts	saveGraph
electron/knowledge-graph-runner.cjs	packages/graph-core/src/persistence/index.ts	saveMeta
electron/llm.cjs	packages/graph-core/src/plugins/extractors/__tests__/php-extractor.test.ts	process
electron/main.cjs	electron/gh-path.cjs	augmentPathForCli
electron/main.cjs	electron/compare-runner.cjs	runCompareEngineAsync

From file	To file	Callee
electron/main.cjs	electron/compare-runner.cjs	runCompareEngine
electron/main.cjs	electron/llm.cjs	summarizeChange
electron/main.cjs	electron/llm.cjs	summarizeChangeStream
electron/main.cjs	electron/llm.cjs	probeProvider
electron/main.cjs	electron/prefetch-summaries.cjs	stopSummaryPrefetchWorker
electron/main.cjs	electron/prefetch-summaries.cjs	clearSummaryPrefetchCache
electron/main.cjs	electron/prefetch-summaries.cjs	startSummaryPrefetchWorker
electron/main.cjs	electron/prefetch-summaries.cjs	getPrefetchedSummary
electron/main.cjs	electron/file-summary-export.cjs	exportFileSummaryArtifacts
electron/main.cjs	electron/file-summary-export.cjs	readFileSummaryMarkdowns
electron/main.cjs	electron/file-summary-export.cjs	exportSelectionSummaryArtifacts
electron/main.cjs	electron/file-summary-export.cjs	readSelectionSummaryMarkdowns
electron/main.cjs	electron/code-city-model.cjs	buildCodeCityModelPayload
electron/main.cjs	electron/novadiff-docs-writer.cjs	writeNovadiffDocsBundle
electron/main.cjs	electron/novadiff-docs-html.cjs	writeNovadiffDocsHtml
electron/main.cjs	electron/novadiff-docs-pdf.cjs	generateNovadiffDocsPdf
electron/main.cjs	electron/novadiff-docs-html.cjs	refreshNovadiffDocsHtmlFromDisk
electron/main.cjs	electron/knowledge-graph-runner.cjs	buildKnowledgeGraph
electron/main.cjs	electron/git-service.cjs	detectGitTooling
electron/main.cjs	electron/github-service.cjs	getAuthStatus
electron/main.cjs	electron/git-service.cjs	getRepoStatus
electron/main.cjs	electron/repo-discovery.cjs	discoverRepos
electron/main.cjs	electron/repo-discovery.cjs	matchRepoForGithubRepo
electron/main.cjs	electron/github-service.cjs	listRepos
electron/main.cjs	electron/github-service.cjs	listPullRequests
electron/main.cjs	electron/git-service.cjs	isGitRepo
electron/main.cjs	electron/git-service.cjs	preparePrCompareRoots
electron/main.cjs	electron/github-service.cjs	viewPullRequest
electron/main.cjs	electron/git-publish.cjs	previewPublish
electron/main.cjs	electron/git-publish.cjs	executePublish
electron/main.cjs	electron/git-service.cjs	stagePaths
electron/main.cjs	electron/export-snapshot.cjs	exportProjectSnapshotZip
electron/main.cjs	electron/git-service.cjs	stageDistrict
electron/main.cjs	electron/workspace-store.cjs	loadSession
electron/main.cjs	electron/github-service.cjs	listDetectedAccounts
electron/main.cjs	electron/github-service.cjs	getUserProfile
electron/main.cjs	electron/gh-install.cjs	getGhToolingStatus
electron/main.cjs	electron/gh-install.cjs	installGh
electron/main.cjs	electron/workspace-store.cjs	setLocalOnlyMode
electron/main.cjs	electron/github-auth-flow.cjs	startGithubDeviceAuth
electron/main.cjs	electron/github-auth-flow.cjs	cancelGithubDeviceAuth
electron/main.cjs	electron/workspace-store.cjs	setGitUser
electron/main.cjs	electron/workspace-store.cjs	workspacesRoot
electron/main.cjs	electron/git-service.cjs	parseGithubSlugFromUrl
electron/main.cjs	electron/workspace-store.cjs	createWorkspace
electron/main.cjs	electron/workspace-history.cjs	indexWorkspaceHistory
electron/main.cjs	electron/workspace-store.cjs	getWorkspace
electron/main.cjs	electron/workspace-store.cjs	setActiveWorkspace
electron/main.cjs	electron/workspace-store.cjs	listWorkspaces
electron/main.cjs	electron/workspace-store.cjs	updateWorkspaceLiveRepo
electron/main.cjs	electron/workspace-store.cjs	updateWorkspaceUiState
electron/main.cjs	electron/git-blame.cjs	blameFileAtRef
electron/main.cjs	electron/git-commit.cjs	getCommitDetail
electron/main.cjs	electron/github-commit-context.cjs	getGithubCommitContext

Codebase outline (JSON excerpt)

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    {
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    },
    {
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  {
    "count": 1,
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  }
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    "via": "blameFileOwnership"
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    "via": "htmlToPdfFile"
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    "to_file": "electron/gh-path.cjs",
    "via": "augmentPathForCli"
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  {
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```

```
"via": "isGhInstalled"
},
{
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  "to_file": "electron/gh-path.cjs",
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{
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},
{
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  "via": "createPullRequest"
},
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  "to_file": "electron/gh-path.cjs",
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},
{
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  "to_file": "electron/gh-path.cjs",
  "via": "resolveGhExecutable"
},
},
```

```

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  "to_file": "electron/gh-path.cjs",
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  "to_file": "electron/github-service.cjs",
  "via": "getAuthStatus"
},
{
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  "to_file": "electron/github-service.cjs",
  "via": "tryRunGh"
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  "to_file": "electron/gh-path.cjs",
  "via": "resolveGhExecutable"
},
{
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  "to_file": "electron/gh-path.cjs",
  "via": "augmentPathForCli"
},
{
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  "to_file": "electron/gh-path.cjs",
  "via": "ghNotFoundError"
},
{
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  "to_file": "electron/git-service.cjs",
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},
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  "to_file": "packages/graph-core/src/plugins/parsers/index.ts",
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},
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  "to_file": "packages/graph-core/src/analyzer/graph-builder.ts",
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},
{
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```

```
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  },
  {
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    "to_file": "electron/llm.cjs",
    "via": "probeProvider"
  },
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    "via": "stopSummaryPrefetchWorker"
  },
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    "via": "clearSummaryPrefetchCache"
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  },
  {
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  },
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  },
  {
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    "to_file": "electron/file-summary-export.cjs",
    "via": "readSelectionSummaryMarkdowns"
  },
  {
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    "to_file": "electron/code-city-model.cjs",
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    "via": "writeNovadiffDocsBundle"
  },
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    "via": "generateNovadiffDocsPdf"
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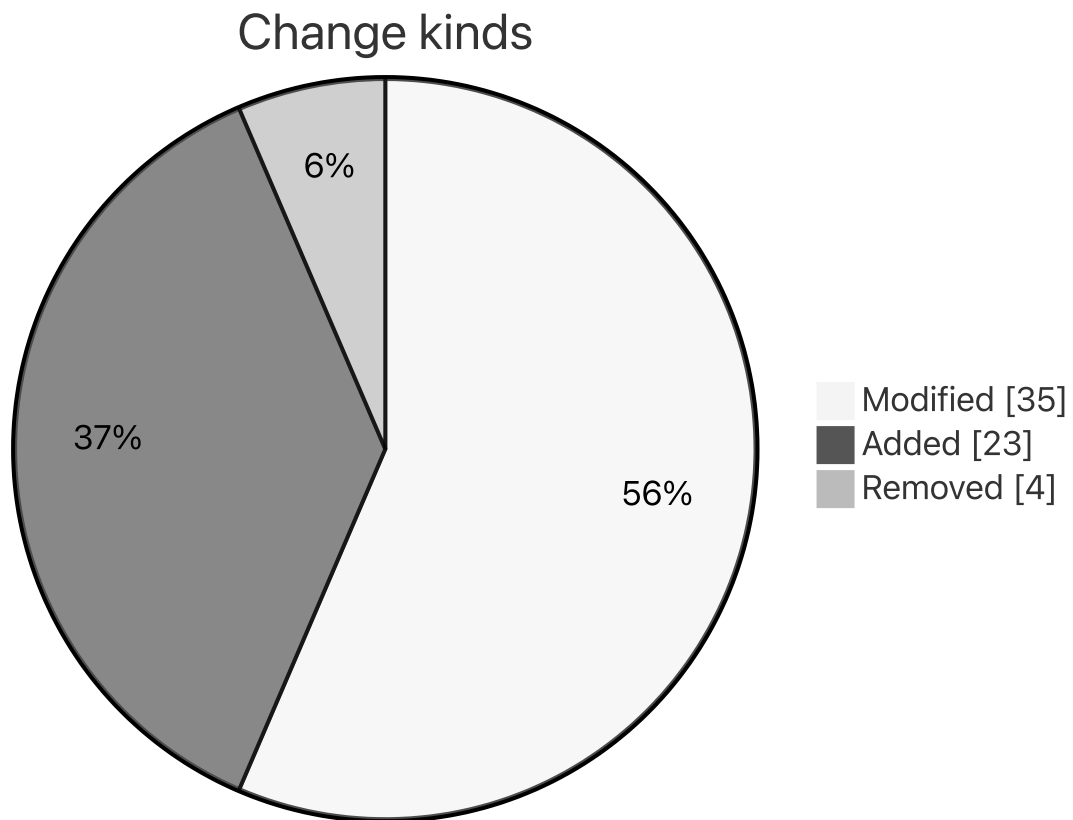
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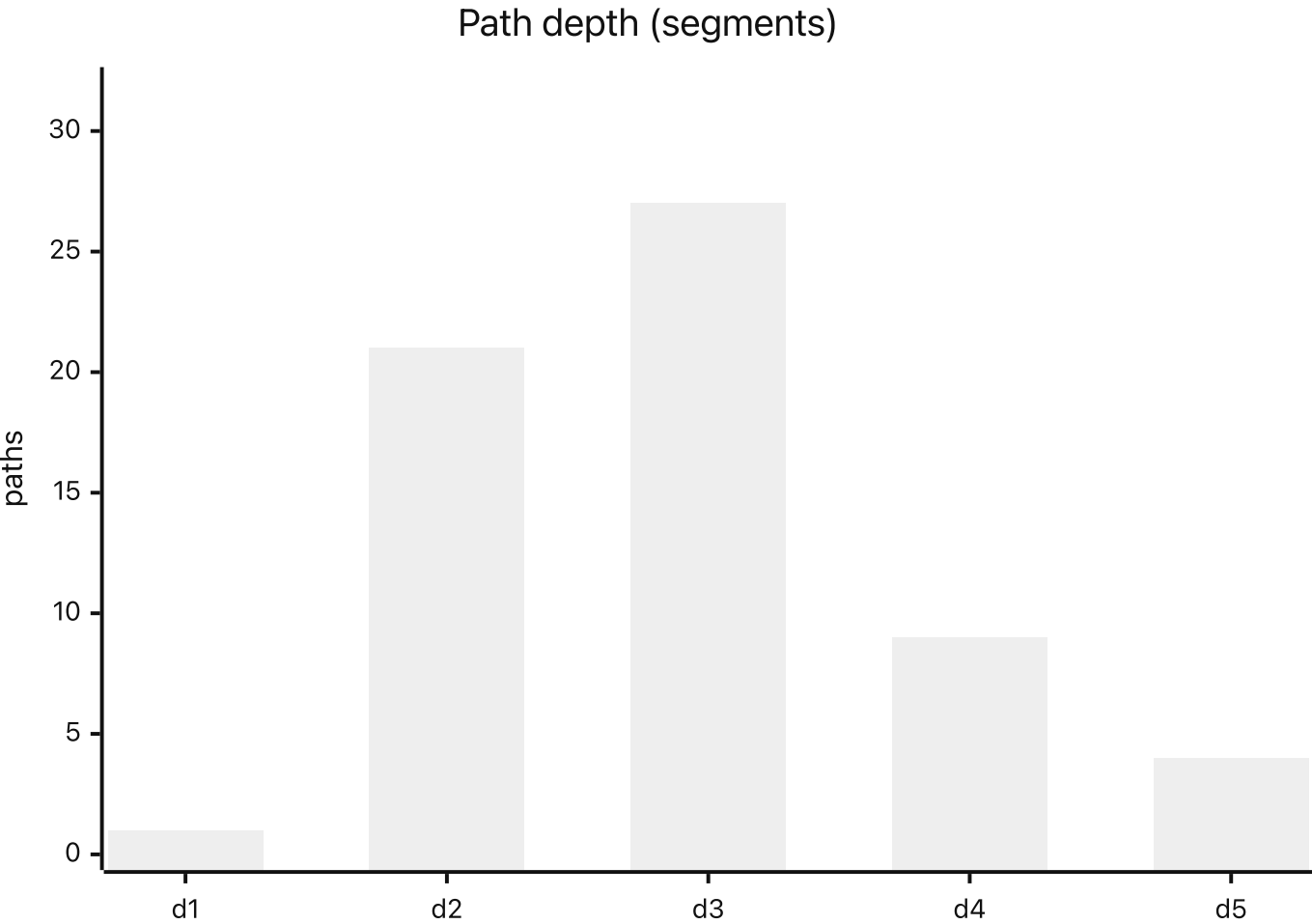
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  "via": "discoverRepos"
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{
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{
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  "via": "exportProjectSnapshotZip"
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  "to_file": "electron/git-service.cjs",
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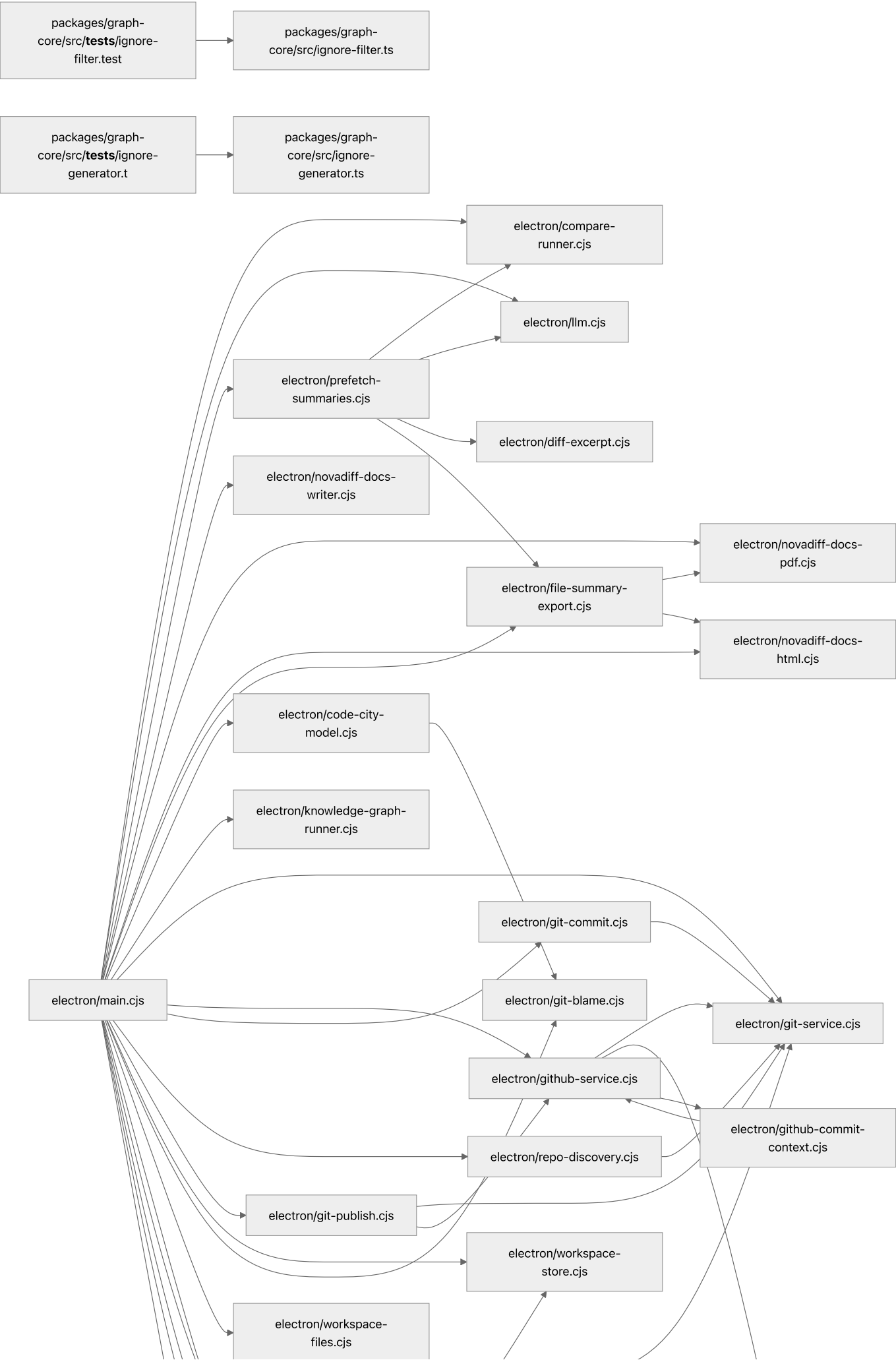
Diagrams (Mermaid)

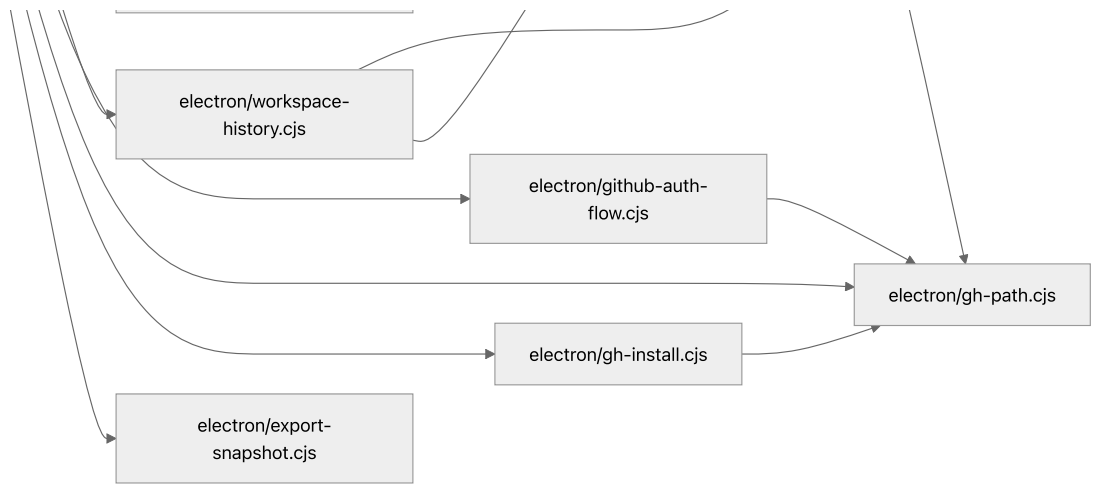
Change mix



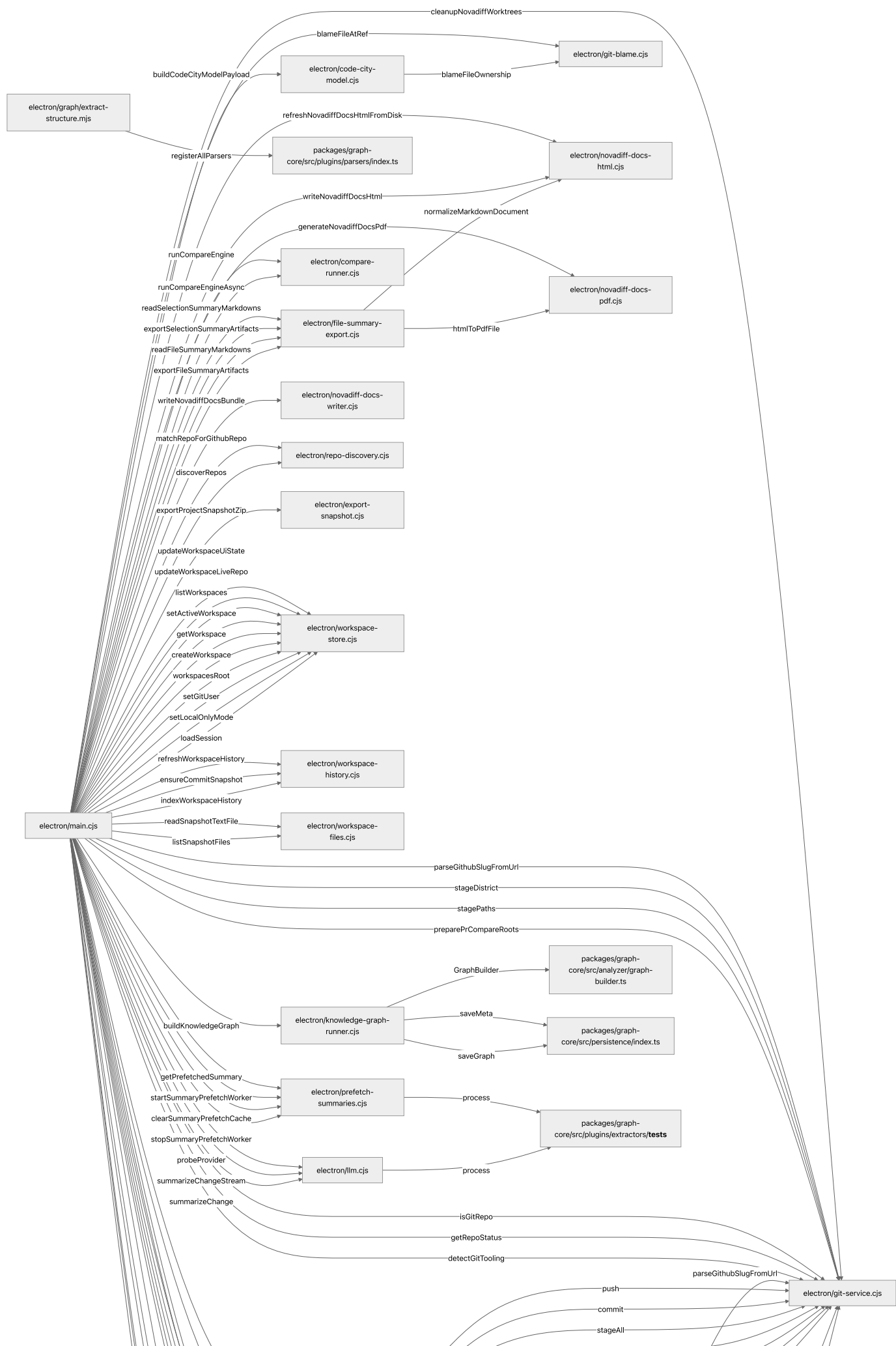


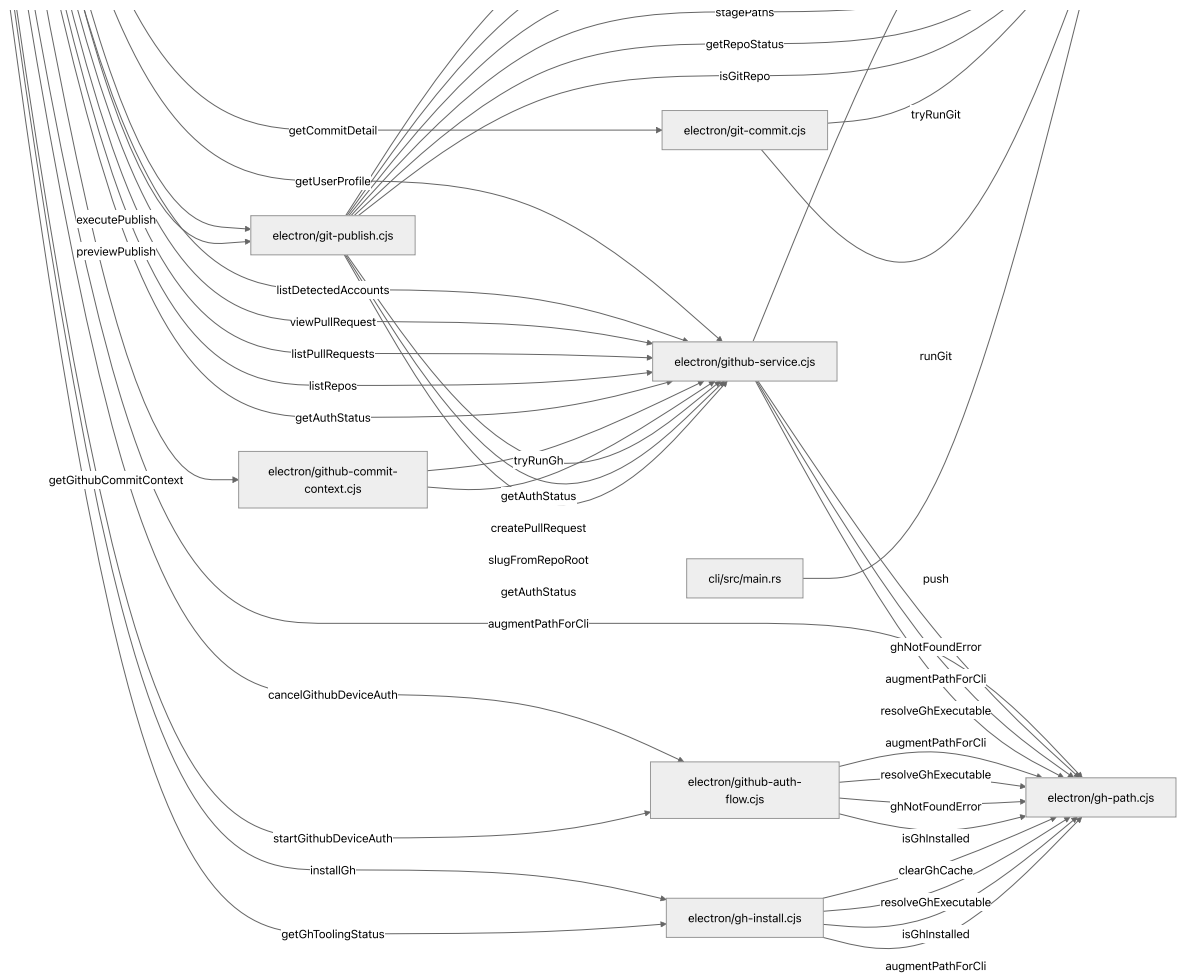
Module import graph (relative imports)





Cross-file calls (heuristic)





Types / classes (regex sketch)

CDrawExtractor	ChangeAnalysis	ChangeKind	ClassFingerprint	ClassExtractor	DocuRefParser	CrossedEdges	DocParser	FileAnalysisMeta	FileChangeResult	FileFingerprint	FileMeta	FingerprintState	FrameworkRegistry	FunctionFingerprint	GenetExtractor	GetExtractor	GraphBuilder	GraphGLParser	IgnoreFile	ImportInspector	JSONConfigParser	JsonExtractor	LLMFileAnalysis
----------------	----------------	------------	------------------	----------------	---------------	--------------	-----------	------------------	------------------	-----------------	----------	------------------	-------------------	---------------------	----------------	--------------	--------------	---------------	------------	-----------------	------------------	---------------	-----------------